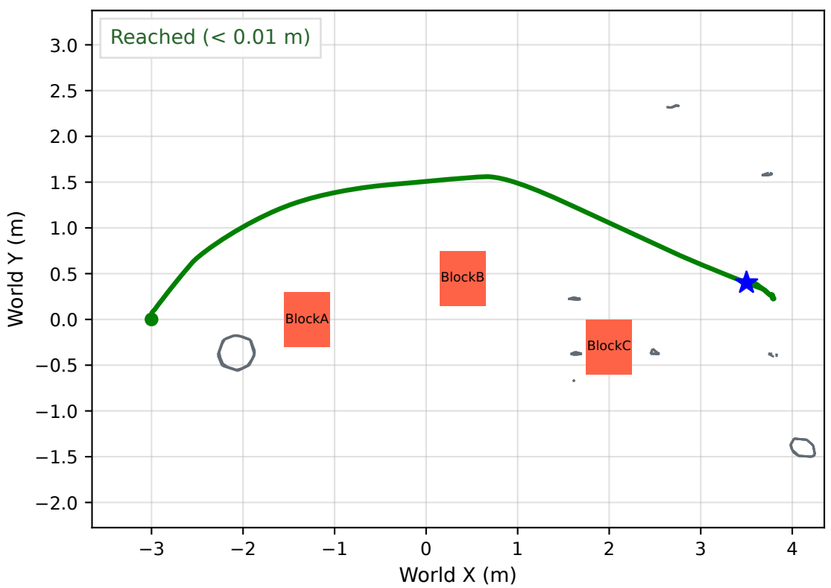
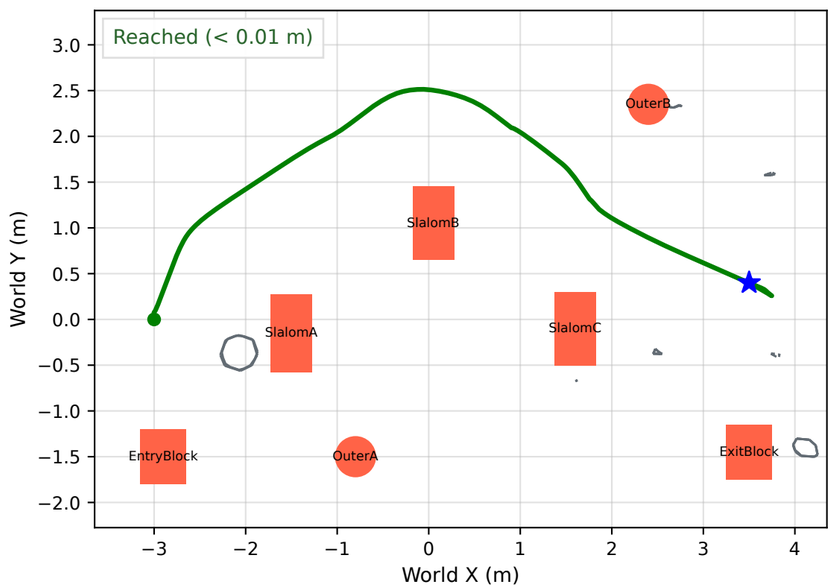
ROV obstacle-avoidance routes across five pool scenarios

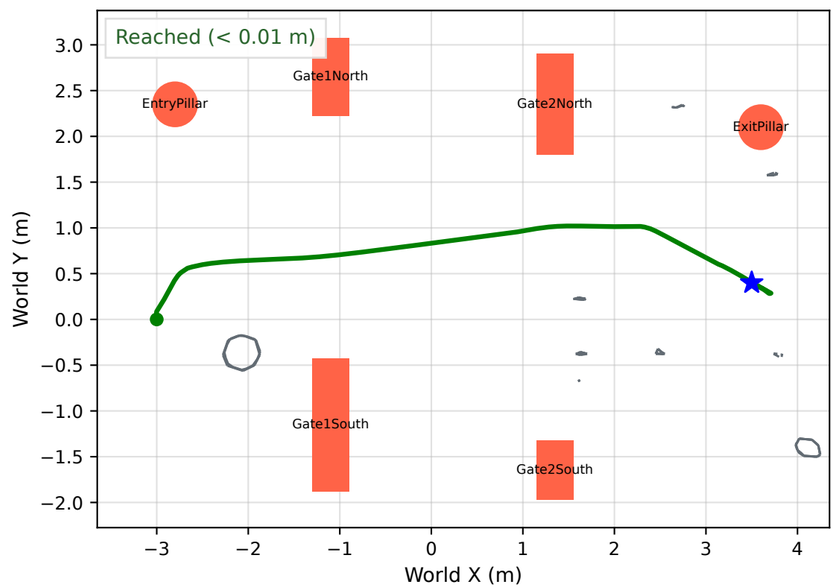
Baseline | goal error: 0.0034 m



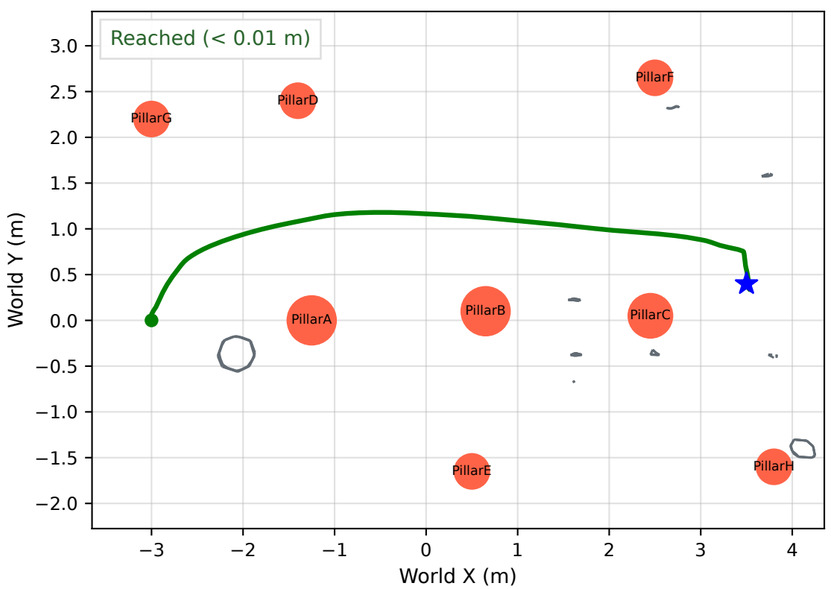
Slalom | goal error: 0.0029 m



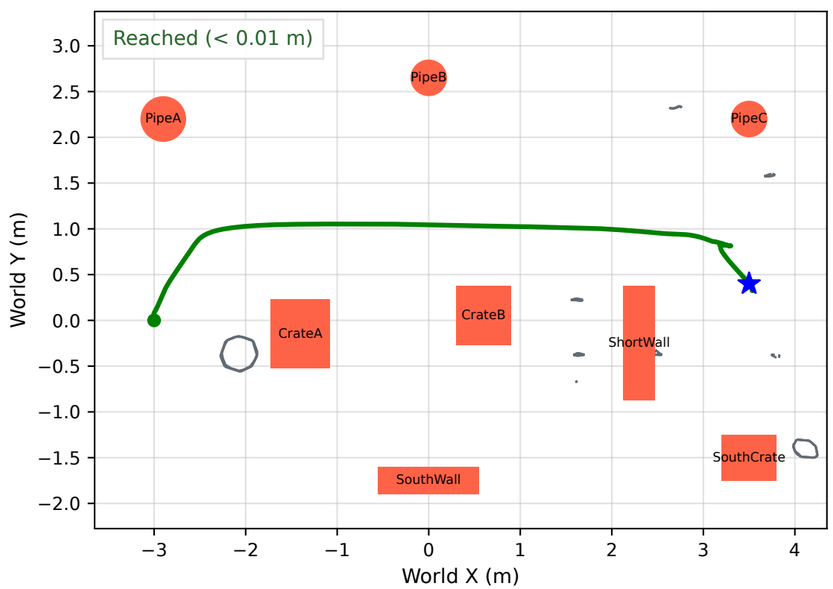
Offset gates | goal error: 0.0015 m



Pillar array | goal error: 0.0009 m



Mixed obstacles | goal error: 0.0012 m



Executed ROV path
Start
Goal
Added obstacle footprint
Original pool structures
Final position: not reached

Measured navigation trajectories
Start: [-3.0, 0.0, 1.0] m
Goal: [3.5, 0.4, 1.0] m
Fixed reference height: z = 1.0 m
HFOV: 90 deg; depth stride: 1
Ideal depth + ground-truth odometry
Arrival tolerance: 0.01 m; dwell: 2 s
Simulation budget: 180 s per scenario

Routes are plotted without smoothing
or extending them to the goal.